Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “separation” to explain your answer. Sentence starters are optional.

gravity: an attractive interaction between masses

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A **3 kg** object moves at **5 m/s** in an ideal circle under a force perpendicular to velocity. **Explain which parts of velocity** change and whether the force does work.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes **$1/2^2$=1/4** as large. A **20 N** interaction would become **5 N**.

Your turn: 1. **Sketch the two separations** using centers.
2. **Draw equal** and opposite force arrows for each pair.

Sentence starter: In my work, I used … to show … . I labeled it … .

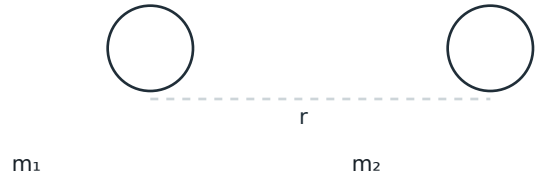


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal spheres are shown. **Mark the distance r used** by the gravity equation and **draw a force** on each sphere.

Sentence starter: In my work, I used … to show … . I labeled it … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulator begins with attraction F_0. Multiply the first mass by **3**, the second by **2**, and the center-to-center distance by **3**. **Complete the diagram of multiplying** factors and **find F**/F_0.

Sentence starter: I found … [unit]. This value means … in this situation.

Factor in $F = Gm_1m_2/r^2$	Multiplier
m_1	3
m_2	2
r^2	____

Marzano check - circle: ① ② ③

Universal gravitation

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does a large planet exert more force on a small moon than the moon exerts on the planet? **Compare forces** and accelerations.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A spacecraft of mass **4 kg** pushes an asteroid of mass **400 kg** with **24 N**. **Compare interaction forces** and the accelerations due to that interaction.
Put each partner force on the object it acts on.

Word help: interaction = two objects affecting one another

Sentence starter: Both … . They differ in … because … .



Object A



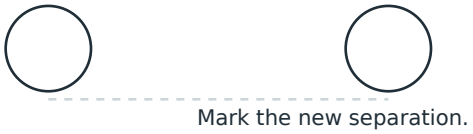
Object B

Marzano check - circle: 1 2 3

07 Exit check

What separation makes force **1/9** of its original value? What if both masses also triple?

Sentence starter: I notice … . This supports … because … .



Mark the new separation.

Mass 1 Mass 2

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Both masses triple. What separation multiplier keeps the force unchanged? What multiplier makes it half the original force?

Sentence starter: I notice … . This supports … because … .

Quantity	Original	New
m_1	m_1	____
m_2	m_2	____
r	r	____
Force	F_0	____

Marzano check - circle: 1 2 3